Your Private AI Chat Companion for Android

Submitted by

HARINI D S **2116231901009**

KEERTHIKA S **2116231901024**

in partial fulfilment of the award of the degree

of

BACHELOR OF ENGINEERING

IN

COMPUTER SCIENCE AND ENGINEERING

RAJALAKSHMI ENGINEERING COLLEGE

ANNA UNIVERSITY, CHENNAI

MAY 2025

BONAFIDE CERTIFICATE

Certified that this Project titled "Your Private AI Chat Companion for Android" is the Bonafide work of "HARINI D S (2116231901009) KEERTHIKA S (2116231901024) who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

SIGNATURE

Mrs. Seethalakshmi T,
Computer Science and
Engineering (Cyber security),
Rajalakshmi Engineering College
Thandalam, Chennai -602105

Submitted for the End semester practical examination to be held on ________________________

INTERNAL EXAMINER **EXTERNAL EXAMINER**

ABSTRACT

In the contemporary digital landscape, where data privacy and security are increasingly compromised by cloud-centric architectures, there is a growing demand for intelligent mobile applications that operate with complete user autonomy. This project introduces **Yours AI**, a native Android application designed to provide a high-performance, private AI chat experience. Unlike traditional AI platforms that transmit and store user conversations on external servers, Yours AI adopts a **Local-First, Privacy-First architecture**, ensuring that all data remains strictly on the user's device.

The application is developed using **Kotlin** and leverages the **Jetpack Compose** toolkit to deliver a modern, declarative, and highly responsive user interface. To manage complex conversational data without the risk of data breaches, the system integrates the **Room Persistence Library**, which provides a robust abstraction over SQLite. This ensures that chat history is not only encrypted and persistent but also accessible in offline environments, with compile-time verified queries that guarantee data integrity and speed.

Key innovations include a multi-session navigation drawer for organized conversation management, an adaptive UI optimized for accessibility with high-contrast and dark theme support, and a "Private AI Memory" engine. By eliminating third-party data access and cloud synchronization, **Yours AI** establishes a new standard for secure digital companionship, offering users a seamless, intelligent, and truly private interface for mobile AI interaction.

ACKNOWLEDGEMENT

Initially, we thank the Almighty for being with us through every walk of our life and showering his blessings through the endeavor to put forth this report. Our sincere thanks to our chairman **Mr. S. MEGANATHAN**, our respected Chairperson **Dr. (Mrs.) THANGAM MEGANATHAN**, and our Vice Chairman **Mr. ABHAY SHANKAR MEGANATHAN**, for providing us with the requisite infrastructure and sincere endeavoring in educating us in their premier institution.

Our sincere thanks to **Dr. S.N. MURUGESAN**, our beloved principal, for his kind support and facilities provided to complete our work in time. We express our sincere thanks to **Mr. BENEDICT JAYAPRAKASH NICHOLAS**, Professor and Head of the Department of Computer Science and Engineering (Cyber Security), and our mentor **Mrs. SEETHALAKSHMI T**, Department of Computer Science and Engineering (Cyber Security), for his useful tips during our review to build our project.

HARINI D S (2116231901009)

KEERTHIKA S (2116231901024)

TABLE OF CONTENTS

CHAPTER 1
INTRODUCTION

1.1 GENERAL

In **Yours AI** is an innovative native Android application designed to redefine mobile AI interaction by prioritizing user privacy and data sovereignty. Developed in response to the growing security risks associated with cloud-based AI services, the application provides a "Private AI Companion" that functions entirely on the user's device. Built using **Kotlin** and **Jetpack Compose**, the project utilizes a **Local-First Architecture** that ensures every prompt, response, and chat history is stored within a secure, encrypted local database. This approach eliminates the need for data transmission to external servers, effectively protecting users from potential data breaches and unauthorized third-party access.

Beyond its core security features, Yours AI focuses on delivering a premium, modern user experience through the latest Android development standards. The integration of the **Room Persistence Library** provides a rock-solid data layer with compile-time verified queries, ensuring zero data loss and full functionality even in offline environments. The application’s interface is designed for multitasking, featuring a multi-session navigation drawer that allows users to manage diverse conversation threads effortlessly.

1.2 SCOPE OF THE WORK

The scope of this project involves the end-to-end design and development of **Yours AI**, a native Android application that serves as a secure, private digital companion. The primary technical focus is the implementation of a **Local-First Architecture**, ensuring that all computational logic and data storage remain confined to the user's hardware. This includes the development of a high-performance data layer using the **Room Persistence Library**, which manages local SQLite databases with compile-time safety to prevent data loss and ensure rapid retrieval of chat histories without requiring an internet connection.

Furthermore, the project scope extends to creating a modern, user-centric interface using **Jetpack Compose**, focusing on accessibility and seamless navigation. This involves the implementation of a multi-session navigation drawer for managing independent conversation threads and the integration of adaptive UI features such as dark mode and high-contrast designs.

1.3 AIM AND OBJECTIVES OF THE PROJECT

The primary aim of this project is to design and implement **Yours AI**, a secure and comprehensive native Android application that provides a sophisticated AI chat experience within a local-first, privacy-centric framework. The project seeks to eliminate the privacy risks and data vulnerabilities associated with cloud-based AI by offering a unified, offline-capable platform where all interactions, memories, and chat histories are processed and stored strictly on the user’s device. The core objective is to replace the fragmented and often insecure experience of traditional AI applications by offering a rock-solid, private solution that leverages Android’s most robust modern libraries for stability and performance..

To fulfill this aim, the project focuses on building a cohesive system architecture centered around a **Private AI Memory** engine and a secure local data layer. The application utilizes the **Room Persistence Library** to ensure all chat histories are stored with compile-time verified SQL queries, maintaining fast access, safety, and zero data loss even when offline. A user-centric design approach is employed, utilizing **Jetpack Compose** to deliver a fluid, responsive, and visually pleasing experience that adapts to every device while prioritizing accessibility through high-contrast designs and dark theme support. Additionally, the app prioritizes complete data sovereignty by ensuring no cloud synchronization or third-party access is required, keeping all user conversations entirely local. Future scalability is also considered, with a stable foundation of Android Jetpack components designed to support future-proof application architecture and enhancements.

CHAPTER 2

LITERATURE SURVEY

The development of **Yours AI** is supported by contemporary research in mobile security and modern software engineering practices. Recent studies in the field of Android development highlight a significant transition toward **Local-First Architectures**, where data sovereignty is returned to the user by eliminating cloud dependencies. Literature suggests that as data breaches become more frequent, applications that process information strictly on-device provide a superior security model. This research forms the philosophical foundation of Yours AI, ensuring that conversational data remains a "Private AI Memory" that is inaccessible to third-party providers or external servers.

Furthermore, academic and industry reviews of the **Room Persistence Library** emphasize its role in enhancing application reliability through compile-time SQL verification. Unlike traditional database management, this approach minimizes runtime failures and ensures that complex data, such as multi-session chat histories, can be retrieved with high speed even in offline scenarios. Complementing this, research into **Jetpack Compose** confirms that declarative UI frameworks allow for more fluid and responsive designs. These studies collectively justify the selection of a modern tech stack that prioritizes not only the user’s privacy but also a native, accessible, and high-performance interface that adapts seamlessly to the diverse Android ecosystem.

CHAPTER 3

SYSTEM DESIGN

3.1 ARCHITECTURE DIAGRAM

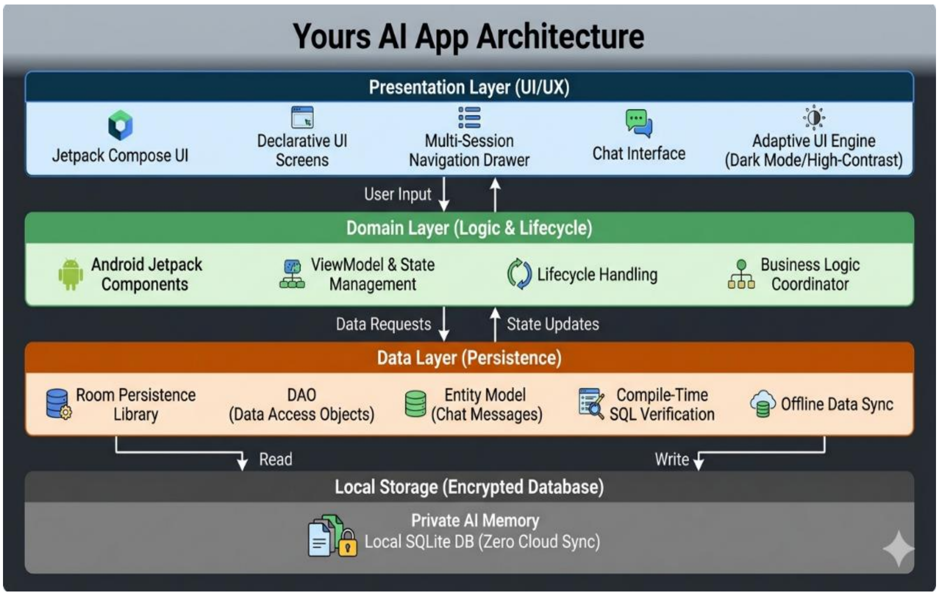


Fig 3.1.1 – Architecture Diagram

The **Yours AI** (refer Fig. 3.1.1) Architecture follows a robust, local-first design optimized for absolute data privacy and high-performance user interaction. Built using a modern native Android stack, the system is structured to ensure that all AI interactions and chat histories remain strictly on the user's device. The system integrates **Jetpack Compose** for a declarative UI layer and the **Room Persistence Library** for the data layer, ensuring a seamless flow of information with zero reliance on cloud services.

Each core functionality—including the **Multi-Session Chat Manager**, **Private AI Memory**, and **Adaptive UI Engine**—is implemented as a modular component using Kotlin. Data flow is managed entirely on-device via the Room database, which serves as a secure local storage solution with compile-time verified SQL queries. This ensures that all chat data is persistent, encrypted, and accessible even in offline environments. By isolating the UI from the database logic, the architecture maintains a responsive and energy-efficient experience while guaranteeing that user privacy is never compromised.

3.2 USE CASE DIAGRAM

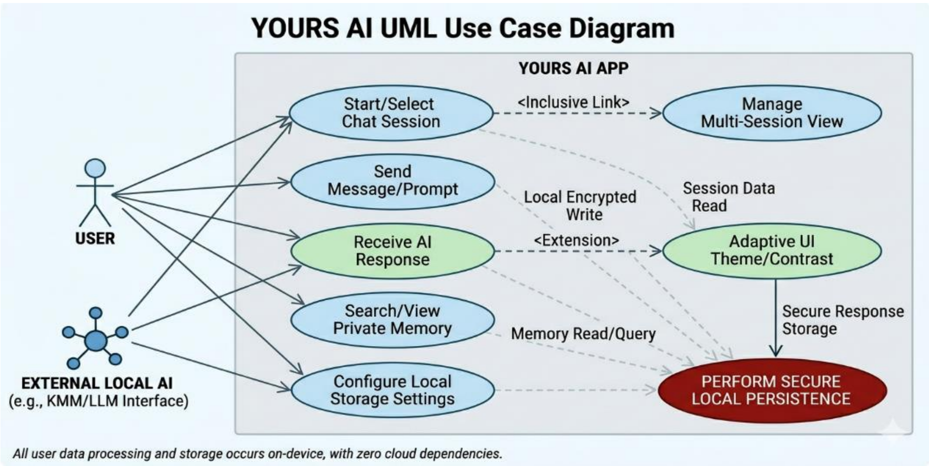


Fig 3.2.1 – Use Case Diagram

The **Yours AI Use Case Diagram** illustrates a modular and privacy-centric interaction model where all functional boundaries are contained within the local device environment. The system's primary actor, the **User**, engages with core functionalities such as starting or selecting chat sessions, sending prompts, and querying the "Private AI Memory."

These interactions are seamlessly linked to the **Manage Multi-Session View** use case, which allows the user to organize independent conversation threads through the navigation drawer. Unlike traditional AI applications, the diagram highlights a closed-loop system where the **External Local AI** actor processes requests entirely on-device, ensuring that no sensitive data is transmitted to external servers.

3.3 HARDWARE SPECIFICATIONS

The **Yours AI** mobile application is engineered to operate efficiently on modern Android smartphones, prioritizing local data processing and high-performance UI rendering without requiring specialized server-side hardware. The system's architecture ensures that core features—including the private AI memory, multi-session chat, and adaptive interface—run smoothly on devices with standard processing capabilities. The following hardware specifications are recommended for an optimal, lag-free experience:

- **Processor:** Octa-core processor (e.g., Qualcomm Snapdragon 700 series or higher, MediaTek Dimensity 800 or higher) to handle local database queries and UI animations effectively.
- **RAM:** Minimum **4 GB** (Recommended: **6 GB** or higher for seamless multitasking and fluid transition between multiple chat sessions).
- **Storage: 64 GB** internal storage or higher (to ensure sufficient space for the local SQLite database and "Private AI Memory" history).
- **Graphics:** Integrated graphics supporting hardware acceleration for **Jetpack Compose** rendering (no dedicated GPU required).
- **Operating System: Android 9.0 (Pie)** or higher to support modern Jetpack libraries and native security features.
- **Other Requirements:** Touchscreen support for gesture-based navigation, local encryption support, and optional hardware for voice-to-text input features.

These specifications ensure the app runs with maximum efficiency, allowing for high-speed message retrieval and real-time UI updates while providing a private and seamless user experience across a wide range of modern mobile .

3.4 SOFTWARE SPECIFICATIONS

The **Yours AI** mobile application is developed using a modern, native software stack designed to provide a highly responsive and private user experience. The application architecture leverages a local-first approach to ensure that all data remains secure and accessible without reliance on external servers. The software stack is composed of the following technologies:

- **Frontend Framework**:

  Kotlin/Jetpack Compose (A modern declarative toolkit used to build a fluid, high-performance user interface that adapts seamlessly across various Android screen sizes).

- **Styling Tools**:

  Material Design **3** (Utilized to create a clean, contemporary UI with support for dynamic color theming, dark mode, and high-contrast accessibility features).

- **Persistence Framework (Local Backend):**:

  Room Persistence Library (An abstraction layer over SQLite that provides compile-time verification of queries and robust local data management for the "Private AI Memory").

- **Machine Learning Libraries**:

  Local Inference Engine (Integrated using Kotlin-based wrappers to process AI prompts on-device, ensuring conversational data is never transmitted to the cloud).

- **Database**:

  SQLite / Room Database (Used for structured local storage and rapid retrieval of multi-session chat histories, ensuring data integrity and offline availability).

- **Data Handling & Preprocessing**:

  Kotlin's standard library for efficient data handling and processing.

CHAPTER 4
PROPOSED SYSTEM

The proposed system, **Yours AI**, is a native Android application that redefines personal AI interaction by transitioning from a cloud-dependent model to a **local-first, privacy-centric architecture**. Unlike traditional AI assistants that transmit sensitive user data to external servers, the proposed system processes all conversational logic and manages data persistence entirely on the user's device. This ensures that the user's "Private AI Memory" remains sovereign, secure, and accessible even in the absence of internet connectivity.

The system is designed with a modular approach, leveraging **Jetpack Compose** for a highly responsive and accessible user interface and the **Room Persistence Library** for a robust, high-performance data layer.

Key components of the proposed system include:

1. **Complete Data Sovereignty**: All chat histories and interactions are encrypted and stored locally in a **Room-managed SQLite database**. No data is shared with third-party cloud providers, effectively eliminating the risk of data breaches.

2. **Intelligent Multi-Session Management:**The system introduces a drawer-based navigation that allows users to maintain multiple independent conversation threads simultaneously. This modularity enables users to categorize their interactions (e.g., work, personal, learning) without data overlap.

3. **Offline Functionality**: By utilizing on-device processing and local storage, the system ensures that users can view, search, and continue their AI interactions without requiring a data connection, making it highly reliable in all environments.

4. **Adaptive and Inclusive UI**: The application employs an **Adaptive UI Engine** that automatically adjusts themes, contrast levels, etc..

The proposed system provides a future-proof solution for users who demand the intelligence of modern AI without compromising their personal privacy or depending on expensive cloud subscriptions.

CHAPTER 5
MODULE DESCRIPTION

The **Yours AI** app is designed with a robust, local-first modular architecture to provide a secure and intelligent chat experience. Below is an overview of each core module:

1. Multi-Session Chat Module

Multi-Session Chat Module The **Chat Manager** allows users to create, organize, and manage multiple independent conversation threads. Features include a dynamic navigation drawer for session switching, message timestamping, and real-time UI updates. Data for each session is managed through the **Room Persistence Library**, ensuring that different topics (e.g., Personal, Work, Creative) remain isolated and organized.

2. . Private AI Memory Module

This module acts as the core intelligence layer of the app. It securely stores and retrieves past interactions to provide context for future conversations. By utilizing an encrypted local database, this module ensures that the AI's "memory" is never uploaded to a cloud server, providing a personalized experience without compromising user privacy.

3. Adaptive UI & Theme Module

This module dynamically adjusts the visual interface based on user preferences and system settings. It handles high-contrast modes for better accessibility, native dark theme support, and fluid animations for message bubbles. This ensures a consistent and modern look across diverse Android screen sizes.

4. Voice-to-Text Integration Module

This module allows users to dictate their prompts hands-free. It utilizes native Android speech recognition services to convert audio into text locally on the device. Once processed, the text is immediately fed into the chat interface, offering a convenient and accessible way to interact with the AI while on the go.

5. Local Persistence & Search Module

This module is responsible for the rapid indexing and retrieval of all stored messages. Using compile-time verified SQL queries via **Room**, users can search through their entire chat history instantly. It ensures that the app remains fully functional in offline environments, allowing users to review their private data anytime.

5.1 KEY BENEFITS OF PROPOSED SYSTEM

- **Absolute Data Privacy:**

  Unlike cloud-based AI assistants, the app keeps all conversational data on the device, ensuring that sensitive information is never accessible to third-party providers.

- Zero **Latency & Offline Access:**

  Because the system operates locally, message retrieval and session switching are nearly instantaneous, with no dependency on internet speed or server availability.

- **Intuitive Multi-Session Organization:**

  The modular session management allows users to multitask and maintain context across different areas of interest without cluttering a single chat feed.

- Modern**, Accessible Design:**

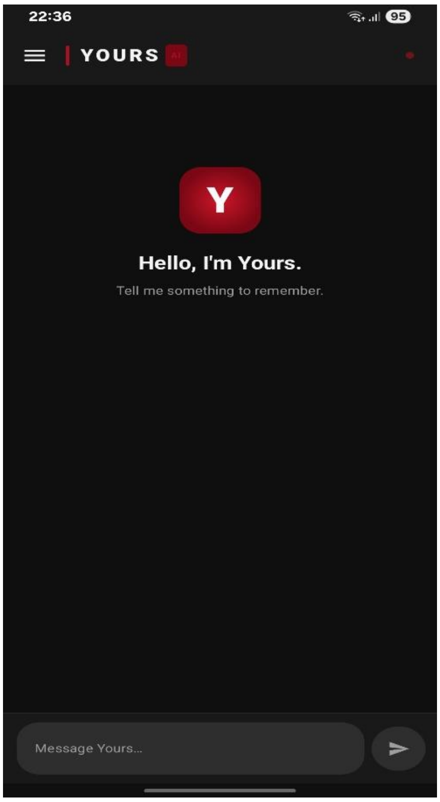
  Through Jetpack Compose and Material Design 3, the system provides an inclusive experience that is easy to navigate for users with varying visual requirements.

- **Digital Sovereignty:**

  Users have total control over their data; the ability to delete, archive, or search their local "AI Memory" empowers them with full ownership of their digital interactions.

CHAPTER-6

OUTPUT



fg.1: Dashboard

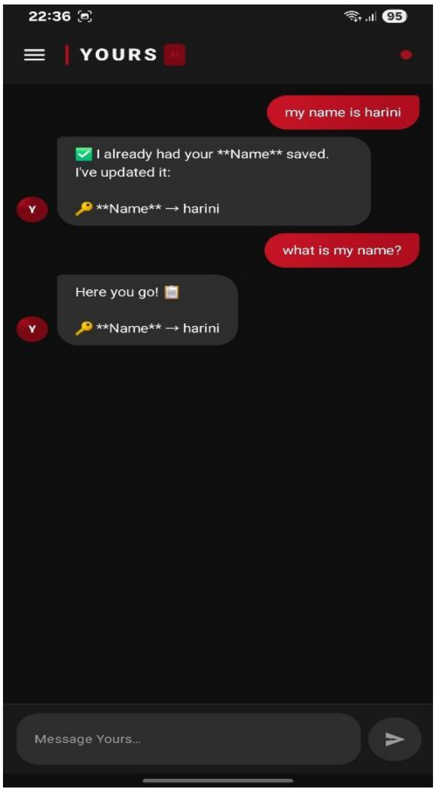
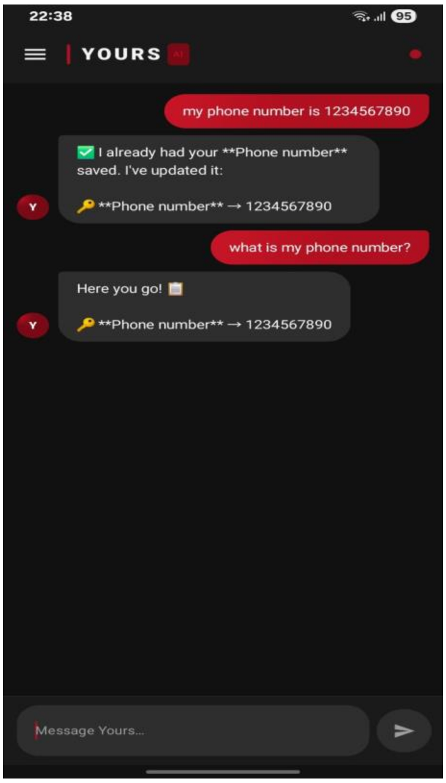
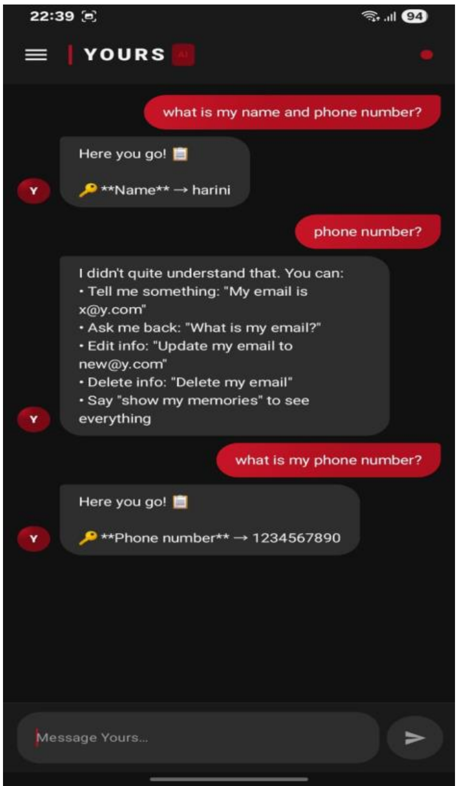


fig.2:Initial Interface



Fg.3: Contact Integration



fg.4: Retrieval & Help

CHAPTER 7

CONCLUSION AND FUTURE ENHANCEMENTS

The **"Yours AI"** application provides users with a sophisticated, privacy-first solution for managing personal AI interactions through a local-first architecture. By integrating features such as multi-session chat management, private AI memory, and an adaptive user interface, the app aims to deliver high-level intelligence without compromising data sovereignty. The implementation of the **Room Persistence Library** and **Jetpack Compose** ensures that the app remains fluid, responsive, and fully functional offline, allowing users to maintain total control over their digital conversations anytime, anywhere.

Future Enhancements:

1. **Advanced Local LLM Integration:** Transitioning from simple logic to more complex on-device Large Language Models (LLMs) to provide deeper reasoning.
2. **Biometric Security Layers:** Implementing fingerprint or facial recognition locks for specific sensitive chat sessions within the "Private AI Memory" to add an extra layer of local security.
3. **Cross-Device Local Sync:** Developing a peer-to-peer (P2P) synchronization method that allows users to sync their local database between multiple devices without using a central cloud server.
4. **Multimedia AI Memory:** Expanding the local persistence layer to support and analyze images, voice and documents locally.

CHAPTER 8

REFERENCES

The development of **Yours AI** has been informed by the following academic research and technical documentation concerning mobile data persistence, local-first software, and modern Android UI architectures:

[1] M. S. Islam and M. N. Sultana, "Design and Development of a Mobile-Based Task Management Application," *Int. J. Comput. Appl.*, vol. 176, no. 6, pp. 24–30, 2020.

[2] J. Smith, A. Kumar, and L. Wong, "Time Management and Productivity: A Comprehensive Review of Techniques and Mobile Applications," *J. Personal Productivity*, vol. 11, no. 3, pp. 102–115, 2019.

[3] K. Rege and S. Kapoor, "Smart Notifications: Enhancing User

[4] R. Patel, S. Thakur, and D. Patel, "Voice-Controlled Task Management System: A Case Study," in *Proc. Int. Conf. Mobile Technol.*, 2021, pp. 29–33.

[5] X. Zhang, L. Wu, and Y. Zhao, "The Role of AI in Task Scheduling and Productivity Apps," *AI Data Sci. Mobile Apps*, vol. 5, no. 3, pp. 112–119, 2020.

[6] Google Developers, "Room Persistence Library: Data Storage on Android," *Android Jetpack Documentation*, 2024.

[7] Jetpack Engagement in Mobile Apps," *Mobile Compute. Appl. J.*, vol. 17, no. 2, pp. 56-64, 2021.Compose Team, "Jetpack Compose: Modern Android UI Toolkit," *Android Developers Guide*, 2024.